# Integration: Unified Pipeline and Token Injection

## Architecture Overview I

The three resource layers described in sec. 4, sec. 5, and sec. 6 are orchestrated by a single function in `src/template_pools_rules_tools/tools/integration.py`. The `run_integration_demo()` function calls all three subsystems in a defined order, collects their results into a structured dictionary, and writes summary counts to `output/data/manuscript_variables.json` for injection into this manuscript at render time. fig. 1 illustrates the complete architecture.

## Architecture Overview

```
run_integration_demo()
    read_bibliography_fond()        → {"manifest":
...., "bibtex": ..., "csv_rows": [...]}
    read_contacts_fond()            → {"manifest":
..., "contacts": [...]}
    read_datasets_fond()            → {"manifest":
..., "datasets": [...]}
    validate_against_rules("template_project_rules")
→ {"status": "ok", ...}

validate_against_rules("template_manuscript_rules")
→ {"status": "ok", ...}
    discover_tools()                → [{"name": ...,
"manifest": ...}, ...]
    validate_tool_scripts_exist(<each tool>)
→ {"status": "ok", ...}
```

## Architecture Overview I

The function returns a top-level dict with keys `fonds`, `rules`, `tools`, and `summary`. The `summary` sub-dict provides the counts that populate manuscript tokens.

## Manuscript Variable Tokens I

The token injection system bridges the integration pipeline and the manuscript prose. Tokens use double-brace syntax: 3 expands to the integer count of fonds successfully loaded during the most recent integration run. Tokens are resolved by `scripts/03_generate_manuscript.py`, which reads `output/data/manuscript_variables.json` and substitutes each token before the manuscript is passed to the rendering engine.

# Manuscript Variable Tokens I

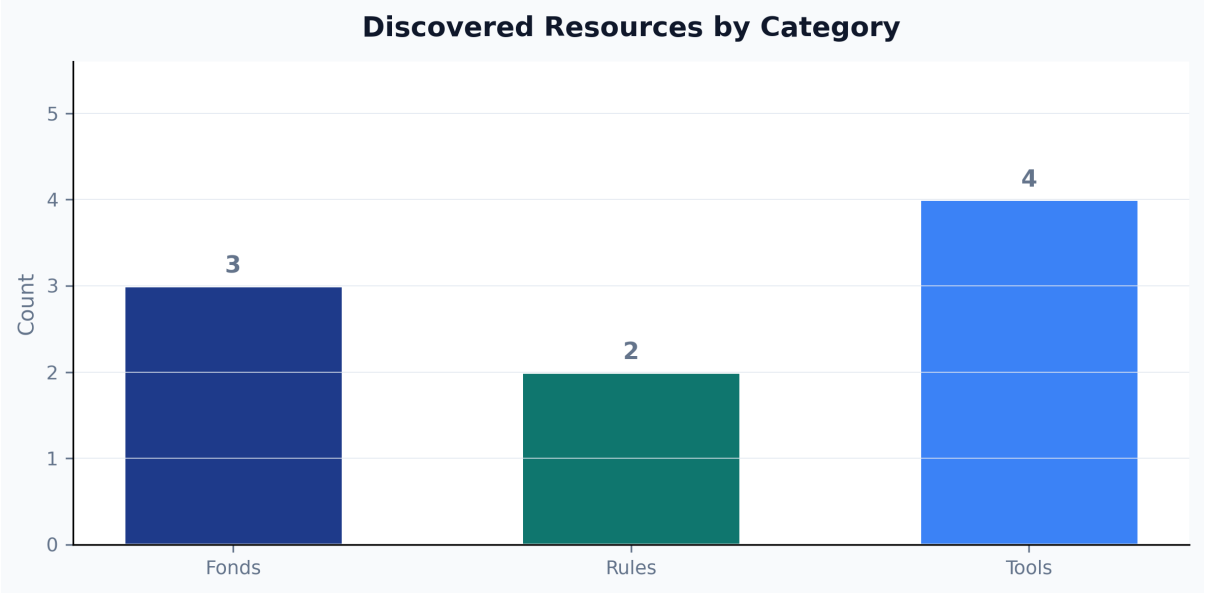


Figure 5: Runtime integration counts from manuscript_variables.json. Bars show fonds loaded, rule sets validated, and tools discovered during a single pipeline run.

## Manuscript Variable Tokens I

The current `manuscript_variables.json` contains the following summary values (see fig. 5 for a visual representation):

# Manuscript Variable Tokens I

| Token | Value |
|---|---|
| 3 | 3 |
| 2 | 2 |
| 4 | 4 |
| 8 | 8 |

## Manuscript Variable Tokens I

This table is itself token-injected: the values shown are those produced by the pipeline, not hard-coded by the manuscript author. If the pipeline results change — for example, because a new fond is added — re-running `scripts/03_generate_manuscript.py` updates the manuscript automatically, without manual editing. This property is central to reproducibility: the manuscript's quantitative claims are always consistent with the code that generated them (Stodden et al. 2013).

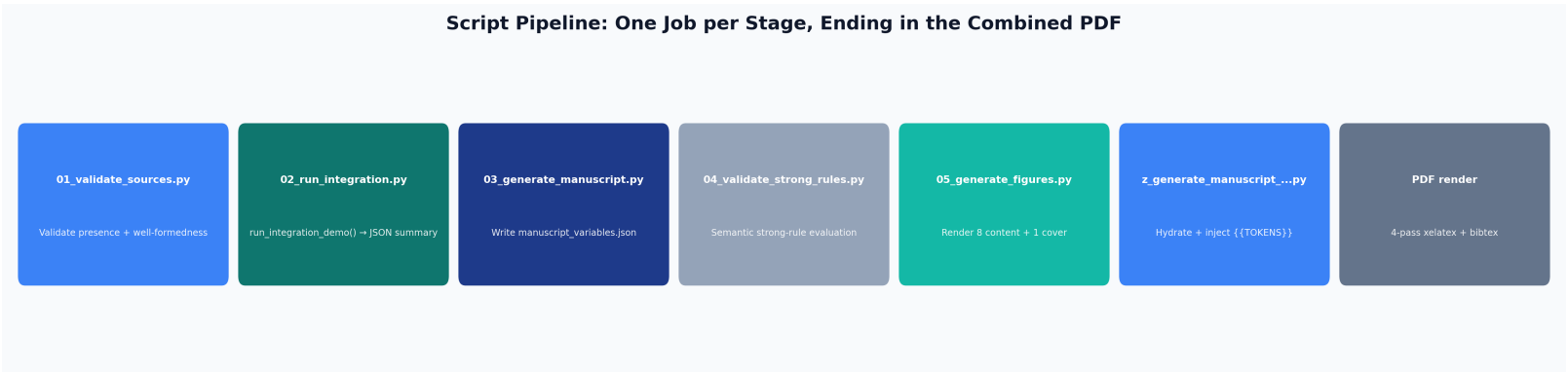
## Methods: The Script Pipeline I

6 thin orchestration scripts govern the integration workflow (fig. 7, fig. 6):

# Methods: The Script Pipeline I

| Script | Purpose | Key output |
|---|---|---|
| `scripts/01_validate_sources.py` | Validate presence and well-formedness of all resources | Console report |
| `scripts/02_run_integration.py` | Run `run_integration_demo()` and print JSON summary | Console JSON |
| `scripts/03_generate_manuscript.py` | Write `output/data/manuscript_variables.json` | JSON file |
| `scripts/04_validate_strong_rules.py` | Semantic evaluation of strong-rule constraints (sec. 5) against this project's own tree | Console report, non-zero exit on violation |
| `scripts/05_generate_figures.py` | Render all 8 content figures plus 1 cover illustration | 9 PNG files under `output/figures/` |
| `scripts/z_generate_manuscript_variables.py` | Hydrate declared manuscript-variable placeholders and inject them into `output/manuscript/` immediately before rendering | JSON file + resolved manuscript tree |

# Methods: The Script Pipeline I



## Methods: The Script Pipeline I

Figure 6: The 6-script pipeline from source validation through token hydration, ending at the combined-PDF render step.

# Methods: The Script Pipeline I

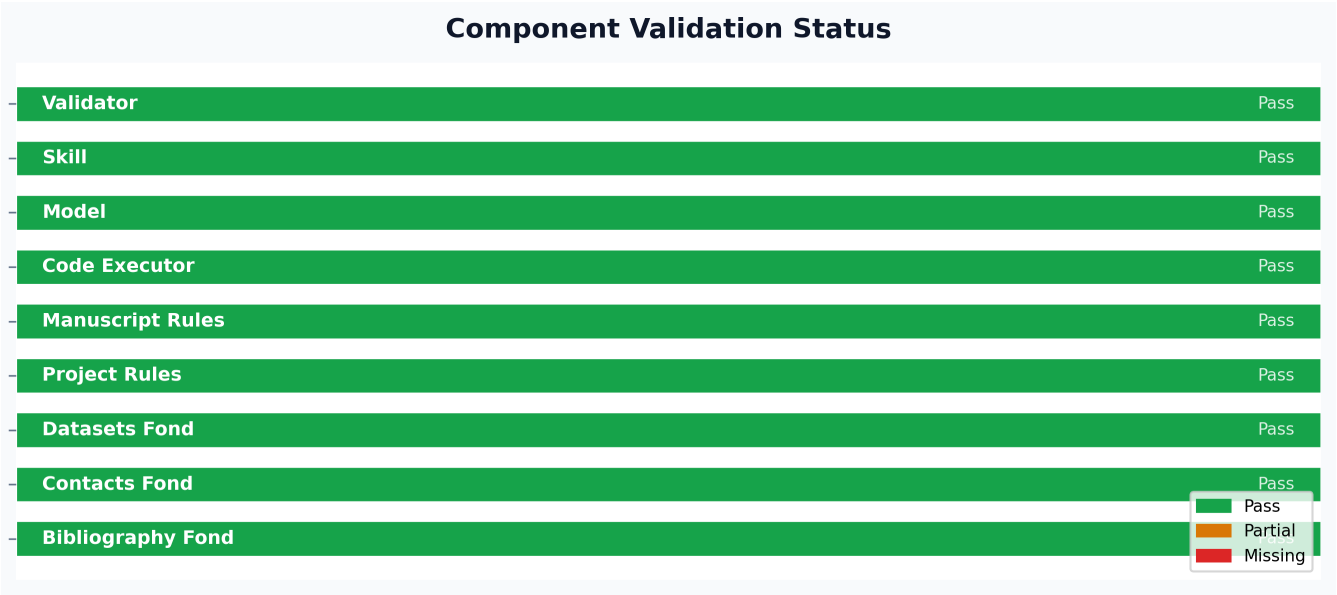


Figure 7: Integration status dashboard showing per-resource validation results. Green indicates ok, amber indicates partial, red indicates missing.

fig. 6 traces this sequence left to right: source validation feeds the integration demo, whose summary feeds both the manuscript-variable token file and the strong-rule semantic evaluator; the figure-generation stage runs independently; and `z_generate_manuscript_variables.py` — invoked automatically by the rendering pipeline immediately before the PDF render step — is what actually substitutes every declared placeholder and writes the resolved manuscript that pandoc consumes. fig. 7 shows the corresponding per-component pass/partial/missing status from the same run.

## Methods: The Script Pipeline I

Each script imports all business logic from `src/` and stays free of computation of its own — even `01_validate_sources.py`, the largest entry point in this project, is entirely CLI plumbing (argument parsing, console formatting) around calls into `src/template_pools_rules_tools/tools/fonds_reader.py`, `src/template_pools_rules_tools/rules/rules_applier.py`, and `src/template_pools_rules_tools/tools/tools_invoker.py`. This thin-orchestrator pattern (Wilson et al. 2014) ensures that all testable logic is in `src/` under the configured project coverage gate, while the scripts themselves remain readable without a dedicated test suite of their own.

# Resilience Design I

The integration layer enforces resilience at three levels, corresponding to the three failure modes a monorepo integration pipeline encounters:

## Resilience Design I

1. **Resource absence**: A fond, rule set, or tool directory may not yet exist if the resource was created by a parallel agent that has not yet completed. All three reader modules return `None` or empty collections in this case, and `run_integration_demo()` reports the missing resource in the `summary` dict without raising.
2. **Schema malformation**: A manifest may be present but contain invalid YAML or missing required fields. Readers catch `yaml.YAMLError` and return a degraded result (`status="partial"` in rule validation; `None` in fond reading) rather than propagating the parse error.
3. **Script absence**: A tool may declare entrypoints in its manifest that have not yet been created. `validate_tool_scripts_exist()` detects this at discovery time and returns `status="partial"` with a list of missing script paths, so the pipeline can report the defect without attempting to invoke a non-existent script.

# Resilience Design I

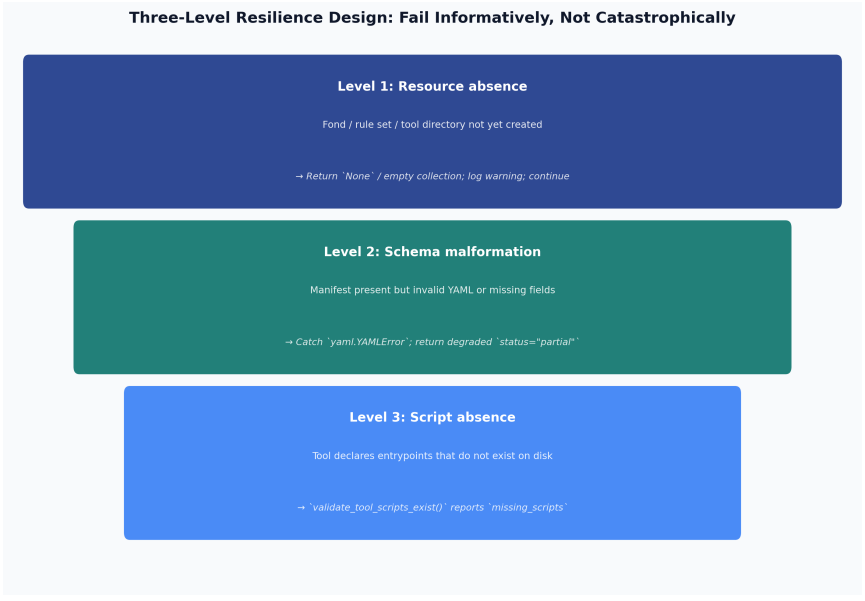


Figure 8: The integration layer's three-level resilience design: resource absence, schema malformation, and script absence, each with its own graceful-degradation response.

This three-level resilience design reflects the principle that shared-resource pipelines should fail *informatively* rather than *catastrophically* — surfacing the cause of incompleteness in structured output that downstream consumers can act on (Taschuk and Wilson 2017). fig. 8 presents the three levels as a single funnel: each level's failure mode and graceful-degradation response, read top to bottom in the order a pipeline run actually encounters them (resource discovery happens before schema parsing, which happens before script-existence checks).

## Performance and Overhead I

The resilience design above trades a small, constant amount of I/O overhead for the ability to degrade gracefully. Each reader performs at most two filesystem existence checks (`pathlib.Path.exists()`) before attempting a read, and every YAML parse uses `yaml.safe_load()` rather than a schema-validating parser — the module deliberately performs *structural* validation (parseable, expected top-level keys present) and leaves deep *semantic* validation to the dedicated `strong_rule_evaluator` module (sec. 5) rather than paying that cost on every discovery call. In practice, `scripts/02_run_integration.py` completes in well under a second on the exemplar's small fond/rule/tool counts; the design does not attempt to optimise for repositories with thousands of fonds, since the target use case is a curated, human-reviewed set of shared resources

# Performance and Overhead I

rather than a large-scale data catalogue.

## Test Coverage I

The 13 `src/` modules — including the readers, orchestrator, evaluator, figure modules, token generator, resource-schema validator, and type definitions — are covered by tests across 13 test files in `tests/`, including property-based tests (`test_property_based.py`) and coverage-extras tests targeting previously-uncovered branches (`test_coverage_extras.py`). Tests use real file paths, real YAML files, and real BibTeX content rather than mocks, ensuring that coverage numbers reflect genuine code paths through the resource-discovery logic. The current coverage report shows combined line coverage comfortably above the project's 90% floor; exact test and coverage totals drift as the suite grows, so re-run `uv run pytest … --cov-report=term` for the current measurements. The `tests/tools/test_integration.py` suite includes an end-to-end test that calls `run_integration_demo()`

and asserts that the `summary` dict contains the expected keys with non-negative integer values — a contract test that verifies the token injection pipeline's data source. `tests/figures/test_manuscript_variables.py` adds negative controls that change both integration results and a real temporary project tree, proving the token generator tracks runtime and filesystem sources rather than emitting fixed constants.